**Which of the statements given above is/are correct?**
(a) 1 and 2 only (b) 3 only
(c) 1, 2 and 3 (d) 2 and 3 only

**Ans.(c):** Liquidity Adjustment Facility (LAF) is a facility extended by RBI to the scheduled commercial banks (excluding RRBs) and Primary dealers (PDs) to avail of liquidity in case of requirement or park excess funds with RBI.
- Only some specified NBFCs are part of PDs and major NBFCs are not part of PDs remain outside the preview of accessing LAF window of RBI.
- Foreign institutional investors (FIIs) who obtain specific approval from SEBI have been permitted to invest 100% of their portfolios in debt securities.
- National stock exchange (NSE) will be the first exchange to offer a separate trading platform for debts.
- Which an aim to develop corporate debt market in the country, SEBI allowed creation of a separate debt segment on stock exchanges.

**53. In India, which of the following can trade in Corporate Bonds and Government Securities?**
**1. Insurance Companies**
**2. Pension Funds**
**3. Retail Investors**
**Select the answer using the code given below:**
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

**Ans.(d):** Government securities (G-sec) is a tradable instrument issued by the central government or the state government.
- Corporate bonds and government securities are two types of investments that companies and governments use to raise money.
- Major players in the G-sec and corporate bond market include commercial bank and Primary dealers (PDs) besides institutional investors like insurance companies.
- Pension funds and Retail investors are main participants in G-Sec and corporate bonds.

**54. Consider the following:**
**1. Exchange-Traded Funds (ETF)**
**2. Motor vehicles**
**3. Currency swap**
**Which of the above is/are considered financial instrument?**
(a) 1 only (b) 2 and 3 only
(c) 1, 2 and 3 (d) 1 and 3 only

**Ans.(d): Financial instruments** are intangible assets, which are expected to provide future benefits in the form of a claim to future cash. It is a tradable asset.
- **Exchange- Trade Funds (ETFs)** are financial instruments that track the performance of a basket of assets (stock, bonds) and allowing investors to buy or sell them.
- **Currency swap** is a financial derivative contract in which two parties exchange the amount of loan in one currency to another currency, based on current exchange rate.
- Motor vehicles are not considered financial instruments. They are tangible assets used for transportation and are not traded as financial securities or derivatives on financial markets.

**55. With reference to the sectors of there Indian economy, consider the following pairs:**

| | Economic activity | Sector |
|---|---|---|
| 1. | Storage of agricultural produce | Secondary |
| 2. | Dairy farm | Primary |
| 3. | Mineral exploration | Tertiary |
| 4. | Weaving cloth | Secondary |

**How many of the above are correctly matched?**
(a) Only one (b) Only two
(c) Only three (d) All four

**Ans.(b):** The economic activity and related sector is as–

| Economic activity | Sector |
|---|---|
| Storage of agricultural produce | Primary sector |
| Dairy farm | Primary sector |
| Mineral exploration | Primary sector |
| Weaving cloth | Secondary sector |

**56. Consider the following materials:**
**1. Agricultural residues**
**2. Corn grain**
**3. Wastewater treatment sludge**
**4. Wood mill waste**
**Which of the above can be used as feedstock for producing Sustainable Aviation Fuel?**
(a) 1 and 2 only (b) 3 and 4 only
(c) 1, 2, 3 and 4 (d) 1, 3 and 4 only

**Ans.(c):** Sustainable Aviation Fuel (SAF) is a type of liquid fuel currently used in commercial aviation (airplanes) to reduce green house gas ($CO_2$) emissions by up to 80%.
- It is made from various source, including waste oil, fats, green and municipal waste and non-food crops.
- Corn grain, oil seeds, Algae, Agricultural residues wood mill waste, wet waste (manure and wastewater treatment sludge) are used to make aviation fuel (SAF).

**57. With reference to physical capital in Indian economy, consider the following pairs:**

| | Items | Category |
|---|---|---|
| 1. | Farmer's plough | Working capital |
| 2. | Computer | Fixed capital |
| 3. | Yarn used by the weaver | Fixed capital |
| 4. | Petrol | Working capital |

**How many of the above are correctly matched?**
(a) Only one (b) Only two
(c) Only three (d) All four

**Ans.(b):** Physical capital in Indian Economy-

| Items | Category |
|---|---|
| Farmer's plough | Working Capital |
| Computer | Fixed Capital |
| Yarn used by the weaver | Working capital |
| Petrol | Working capital |

**58. Which one of the following words/phrases is most appropriately used to deonte ''an interoperable network of 3D virtual worlds that can be accessed simultaneously by millions of users, who can exert property rights over virtual items''?**
(a) Big data analytics (b) Cryptography
(c) Metaverse (d) Virtual matrix

**Ans.(c):** The term **'Metaverse'** refers to an interconnected network of 3D virtual worlds in which users represented by avatars interact.
- The metaverse often conflated (combined) with the virtual reality (VR) and augmented reality (AR) technology used to experience a shared 3D world.
- Users can create and own virtual items within the metaverse.

**59. With reference to the rule/rules imposed by the Reserve Bank of India while treating foreign banks, consider the following statements:**
**1. There is no minimum capital requirement for wholly owned banking subisidiaries in India.**
**2. For wholly owned banking subsidiaries in India, at least 50% of the board members should be Indian nationals.**
**Which of the statements given above is/are correct?**
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2



**Ans.(b):** RBI allows foreign banks to operate in India either through branch presence or they can set up a wholly owned subsidiary (WOS) with near national treatment.
- The initial minimum paid -up voting equity capital far a WOS shall be 5 billion.
- The composition of the board of directors of WOS should meet requirements- not less than 50 percent of the directors should be Indian nationals, as defined under section 10A of the banking Regulation Act, 1949.

**60. With reference to Corporate Social Responsibility (CSR) rules in India, consider the following statements:**
**1. CSR rules specify that expenditures that benefit the company directly or its employees will not be considered as CSR activities.**
**2. CSR rules do not specify minimum spending on CSR activities.**
**Which of the statements given above is/are correct?**
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**Ans.(a): Under the companies (CSR Policy) Rules, 2024,** any activity benefitting employees, of the company shall not be considered as eligible CSR activity.
- As per rule, any activity designed exclusively for the benefit of employees shall be considered an "activity benefitting employees" and will not qualify as permissible CSR expenditure.
- As per rule section 135 of the companies Act 2013, mandates, that certain companies must allocate at least 2% of their average net profits from preceding three financial years towards CSR activity.

**61. With reference to radioisotope thermoelectric generators (RTGs), consider the following statements:**
**1. RTGs are miniature fission reactors.**
**2. RTGs are used for powering the onboard systems of spacecrafts.**
**3. RTGs can use Plutonium-238, which is a by-product of weapons development.**
**Which of the statements given above are correct?**
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

**Ans.(b):** Radioisotope thermoelectric generators (RTGs) are advanced power source developed for the demand of deep space mission that are extraordinarily reliable, due to their longivity and durability in harsh environments.
- RTGs are not miniature fission reactors, they rely on the natural radioactive decay of plutonium-238 to generate using thermocouples.
- RTGs generates electrical power for powering the onboard systems of spacecrafts, especially in situations where solar power is insufficient.
- Plutonium- 238, which is used in RTGs, can be a by-product of nuclear reactors and weapons developments.

**62. Consider the following statements:**
**Statement-I : Giant stars live much longer than dwarf stars.**
**Statement-II: Compared to dwarf stars, giant stars have a greater rate of nuclear reactions.**
**Which one of the following is correct in respect of the above statements?**
(a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-II
(b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
(c) Statement-I is correct, but Statement-II is incorrect
(d) Statement-I is incorrect, but Statement-II is correct

**Ans.(d):** Giant stars have greater rate of the nuclear reactions that light up stars. Massive stars also use up the hydrogen fuel in their core faster, leading to a shorter overall lifespan.
- Having greater masses than sun, these massive stars have hotter and denser cores than dwarf stars, Therefore Giant stars have a greater rate of the nuclear reactions.
- Hence statement-I is incorrect & statement-II is correct.

**63. Which one of the following is synthesised in human body that dilates blood vessels and increases blood flow?**
(a) Nitric oxide (b) Nitrous oxide
(c) Nitrogen dioxide (d) Nitrogen pentoxide

**Ans.(a) :**
- Nitric oxide (Nitrogen Oxide or Nitrogen Monoxide : NO') is a colorless gas.
- Nitric oxide (NO) is produced in the human body from inorganic nitrates in green leafy vegetables, fruits, grains and processed meats. It is stored in the body as nitrate. It is very important for the health of blood vessels.
- It is a vasodilator, which means it relaxes the inner muscles of your blood vessels, causing the vessels to widen. In this way, nitric oxide increases blood flow and lowers blood pressure.
- Nitrous oxide is also called laughing gas. It is also used for clinical anesthesia. The main effect of nitrogen dioxide is on the respiratory system.

**64. Consider the following activities:**
**1. Identification of narcotics on passengers at airports on in aircraft/**
**2. Monitoring of precipitation**
**3. Tracking the migration of animals**
**In how many of the above activities can be radars be used?**
(a) Only one (b) Only two
(c) All three (d) None

**Ans.(c) :** • Heinrich Hertz is considered to be the inventor of RADAR.
- The full form of 'RADAR' is 'Radio Detection and Ranging'.
- It uses radio waves. It determines the range, angle and velocity of objects.
- Radar is used in various fields. Such as identifying narcotics hidden by passengers in airports or planes, by meteorologists in forecasting the weather and monitoring rainfall, in keeping an eye on the migration of animals, in navigation of aircraft, in space and military fields, etc.

**65. Consider the following aircraft:**
**1. Rafale 2. MiG-29 3. Tejas MK-1**
**How many of the above are considered fifth generation fighter aircraft?**
(a) Only one (b) Only two
(c) All three (d) None

**Ans.(d):**
- Rafale and Tejas MK-1 are 4.5$^{th}$ generation fighter aircraft, while MiG-29 is a fourth generation fighter aircraft. Tejas is India's indigenous fighter aircraft.
- The cabinet committee on security has recently approved the prototype development and testing phase of India's indigenous fifth generation stealth fighter jet, 'Advanced Medium Combat Aircraft' (AMCA), to build India's first 5$^{th}$ generation aircraft.
- Some of the major 5$^{th}$ generation aircraft in the world are as follows–
1. Sukhoi SU-57–Russia
2. Lockheed Martin F-35 Lightning II–America
3. Lockheed Martin F-22 Raptor–America
4. Chengdu (AVIC) J-20 (Black Eagle)–China etc.